Transient performance of a brushless LV alternator against ISO 8528-5 G3

Systems-engineering dossier, revision 1.0 (10 Nov 2026). All tables marked "generated" are written from `results/*.json` by `scripts/make_tables.py`; no result number is typed by hand.

Central question. Can a given 800 kVA, 400 V, 50 Hz, 4-pole brushless alternator, with its exciter and AVR, meet the ISO 8528-5 performance class G3 voltage limits when a block load is applied or a large induction motor is started direct-on-line? Which design or AVR parameters decide that?

Short answer. Only just, and not robustly. On FEMM-derived parameters the reference machine meets the G3 block-load dip: −14.7 % against −15 % with the retuned shunt AVR (A1). Once the AVR is retuned it also meets the 4 s recovery (0.97 s). It fails two of the six "shalls":

- the 132 kW direct-on-line motor start, at −22.9 % against −20 % (the largest motor within −20 % is 111 kW);
- with shunt excitation, the sustained fault current, at 0.04 pu against 3 pu.

The deciding parameter is the transient reactance X'd. It is 0.256 pu here. SR-01 allows at most 0.263 pu and the motor start at most 0.214 pu. The AVR, a larger exciter and a PMG change the recovery and the fault current; they move the first dip by no more than 0.3 percentage points. Only the next frame meets every shall, and only together with a PMG (A5).

How to read the evidence. Every engineering value carries a tag:

- **[S]** sourced, with a reference in §12;
- **[D]** derived in this study, with the model and file named;
- **[A]** assumed, a typical order of magnitude.

Every [A] value that could change a conclusion is varied in §8.4. The method follows Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed., Wiley 2016 [W]. Page numbers are the printed ones; the PDF page is the printed page + 30. Other textbooks [T1]–[T5] were not available to re-check equation numbers, so their equations are written out and derived in `docs/design_notes.md` instead of being cited by number (§11).

---

0. Findings

Numbers are from `results/aoa_femm.json`, `results/studies_femm.json`, `results/params_femm.json` and `results/a3_ceiling_femm.json`. All are on FEMM-derived parameters.

1. **FEMM and the analytic design agree within 8 % on everything except the damper.** Xd −8 %, Xq −8 %, X'd −3 %, T'do −5 %, no-load field current +8 % (FEMM vs analytic). X''d is 43 % higher in FEMM: 0.207 against 0.145 pu. The reasons are in §6.1: the analytic cage referral spreads the bars over the whole pole arc, and FEMM resolves the leakage of the bar slits.

2. **The first dip is almost all reactance.** Of the −14.7 % block-load dip, −12.6 points are already there at t = 0^+ with the rotor fluxes frozen. The real excitation adds −2.1 points. The excitation option moves the total only between −14.4 % (A2) and −14.7 % (A1).

3. **The SR-01 margin is 0.3 points and fragile.** Any one of the following breaks it on its own (§8.4): - X'd +20 %; - X''d +20 %; - exciter TE = 0.40 s; - VRMAX = 10 pu; - damper time constants halved, which stands in for real end-ring resistance (FEMM rings are ideal): −15.05 %.

4. **Recovery is an AVR-tuning problem, and it is solved.** The textbook pole-cancellation PID (A0) takes 5.25 s and fails SR-02. The grid-tuned PID recovers in 0.97 s with a 6 % open-circuit step overshoot (DR-03 holds).

5. **The motor start decides the frame.** With the base frame, a 132 kW DOL start dips by −22.3 % (PMG) to −22.9 % (shunt). The largest motor within −20 % is 111–115 kW; on the next frame it is 151–156 kW. To start 132 kW within −20 %, X'd must be at most 0.214 pu (shunt) or 0.221 pu (PMG).
6. **The fault-current shall decides the excitation.** Shunt excitation collapses to 0.04 pu at 1 s. The PMG sustains 2.92 pu, 2.7 % short of 3 pu. An AVR ceiling 4.4 % higher (VRMAX 14.6 instead of 14.0 pu, an assumed value, D5) closes the gap.
7. **The root cause is the reference design's X'd, not the excitation.** A machine with data-sheet-class parameters (X'd = 0.17 pu, midpoints of [S7]) dips only −8.5 % on the same block load and −16.2 % on the 132 kW start, with the same shunt AVR. The study design's X'd of 0.256 pu is above the public range of 0.146–0.19 pu.
8. **A single open rotating diode silently breaks SR-01.** It derates the rectifier to 0.765, and the dip becomes −15.02 %. A shorted diode raises the exciter armature current 2.9× and the dip to −15.67 %. The 1×/6× f_{ex} ratio of the field-frame MMF rises from 0.002 (healthy) to 5.5 (open) and 42 (short), so the failure can be detected from the stationary side.
9. **Frequency is not the limit.** At a 50 % step on the generic turbo-charged engine, the frequency dips by −4.7 % (G3: −7 %) and recovers in 2.5 s (G3: 3 s). The engine adds 0.7–0.9 points to the voltage dip.
10. **Waveform.** The no-load line-to-line THD is 2.27 % with an unskewed stator (SR-07: 5 %). The largest harmonics are the 5th (1.94 %) and the 7th (0.99 %); the 2/3 pitch removes the triplens.

---

1. Need, problem space and stakeholders ([W] ch. 4–5)

System of interest (SOI). The alternator with its excitation system: main machine, brushless exciter, rotating rectifier, AVR and AVR power source.

Higher-order system. The generating set: engine, governor, alternator and controls.

Operating environment. An islanded LV bus with two kinds of load:

- block loads: UPS, IT and HVAC blocks;
- direct-on-line motors: pumps and chillers.

Stakeholders ([W] Principle 4.3, p. 78: the customer and the customer's customer):

Role	Who	What they need from the SOI
User (customer)	Generating-set packager	Certify the set to ISO 8528-5 G3 without oversizing the alternator
End user	Facility operator (e.g. a data-centre standby plant)	Loads ride through block loads and motor starts; downstream protection stays selective during faults
Developer	Alternator design and test engineering	Know which design or AVR parameter decides the answer before a frame is built and bench-tested

Problem statement ([W] Principle 4.15, p. 91). Today an alternator is chosen from catalogue reactances and motor-starting curves. When a set fails the transient test, there are two usual fixes, and both have a cost:

- a bigger frame, which adds material and size;
- re-tuning the AVR on site, which has no traceable basis.

The contributory causes ([W] Principle 4.16, p. 91) are not separated. Machine reactances, excitation forcing, AVR dynamics and engine speed dip all lower the same voltage.

Opportunity. A model chain that is traceable from geometry to the ISO 8528-5 metrics:

1. FEMM field solution;
2. machine parameters;
3. dq dynamics with exciter, AVR and engine.

This chain splits the dip by cause and ranks the fixes on evidence.

Measures ([W] Principles 5.10–5.11, p. 115):

- MOE-1: transient voltage deviation on load application, δU_{dyn}^{-} (ISO 8528-5).
- MOE-2: voltage recovery time t_U.
- MOE-3: the largest DOL motor that can be started within the dip limit.
- MOS-1: relative cost.
- MOS-2: sustained short-circuit current, which keeps protection selective.
- MOS-3: frame size.

2. Concept of operations ([W] ch. 5–6)

Mission phases ([W] Principle 5.7, p. 110):

Phase	Sub-phase	What matters for transient performance
Pre-mission	Factory routine test, type test, commissioning	OCC, field currents and AVR set-up. Bench link: the sister project *alternator-routine-test-station* (§10.3)
Mission	Start and voltage build-up	Shunt excitation builds up from remanence; PMG excitation gives voltage from the first revolution
	Load acceptance	Block steps. The first step is limited by the engine (ISO 8528-5 BMEP-based step)
	Steady supply	±1 % voltage band (G3)
	Motor start	DOL start of the largest pump: low-PF current of about 6× rated for 1–3 s
	Load rejection	Breaker trip of a large block causes overvoltage
	Downstream fault	Three-phase short circuit. Fault current must be sustained long enough for selective tripping
Post-mission	Cool-down, shutdown	Not part of this study

UC-1 "accept block load" ([W] Principles 5.12–5.23, pp. 120–122):

- Actors: generating-set controller, load breaker, AVR.
- Precondition: rated speed and voltage, no load.
- Trigger: the breaker closes on P + jQ.
- Main success scenario: the dip stays at or above −15 %, and the voltage is back within ±1 % within 4 s.

UC-2 "start DOL motor" is the same, with a motor in place of the block. **UC-3 "ride through a downstream fault"**: the fault current stays at or above 3 pu until the downstream breaker clears.

3. Modes and states ([W] ch. 7)

Mode ([W] Principle 7.4, p. 157)	Trigger into the mode (Principle 7.5, p. 159)	Allowable actions	Prohibited actions (Principle 7.10, p. 169)
M0 Off	Stop command	Maintenance, insulation test	Loading
M1 Run-up / build-up	Start; speed above the build-up threshold	—	Loading before the voltage is established
M2 No-load ready	Voltage within ±1 %	Close the load breaker	Paralleling without synchronising
M3 Loaded, steady	Load connected, transients settled	Further steps up to the engine step limit	A step larger than the declared acceptance

Mode ([W] Principle 7.4, p. 157)	Trigger into the mode (Principle 7.5, p. 159)	Allowable actions	Prohibited actions (Principle 7.10, p. 169)
M4 Transient (dynamic state, Principle 7.9, p. 166)	Load step, motor start or rejection	AVR forcing up to the ceiling	A second step before recovery
M5 Fault sustain	Terminal voltage collapsed, high current	Hold excitation (PMG) for the selective trip	Over-excitation beyond its time limit
M6 Protected shutdown	Over-voltage, loss of excitation, diode failure	De-excite, alarm	Automatic restart
T Bench test	Factory test mode	Separate drive, test supplies	—

Configuration states (Principle 7.7, p. 164):

- excitation source: shunt or PMG;
- AVR tuning set;
- frame: base or next size up.

Environmental state (Principle 7.8, p. 165): hot windings. Resistances are taken at 115 °C [A].

4. Requirements ([W] ch. 19–22)

Each requirement has the structure of Principle 22.5 (p. 449): subject, *shall*, action, outcome, condition. Each also has a verification method from the set in Principle 22.4 (p. 448).

Source of the G3 values. They are the ISO 8528-5 performance class operating limits (ISO 8528-5:2018, clause 15; ISO 8528-5:2022, Table 4) [S2]. The numbers were read through two manufacturer white papers that reproduce the table [S5][S6], because the standard is paywalled; only its contents and definitions were seen [S2]. Before any external use, check them against a purchased copy.

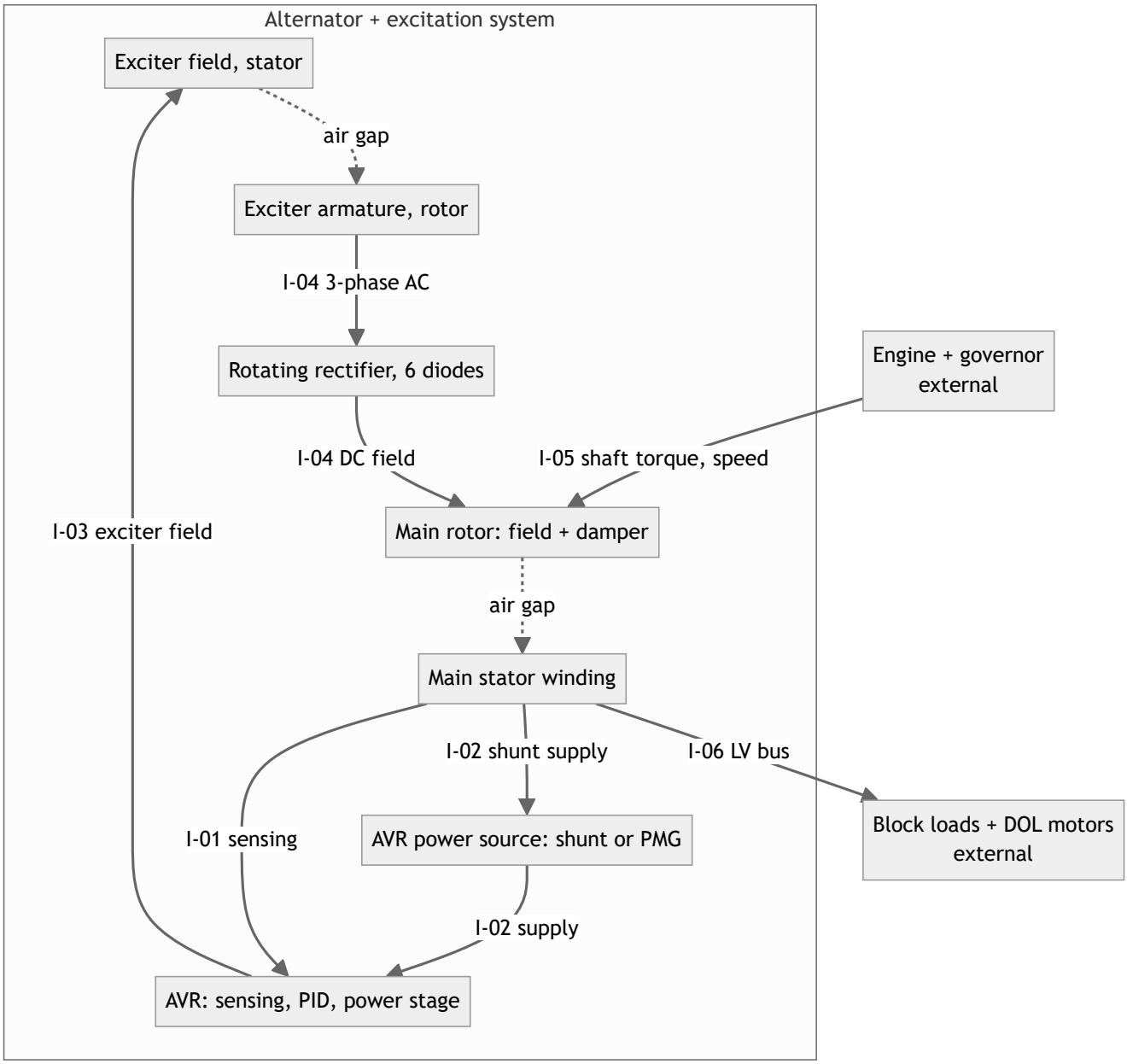
ID	Requirement	Source	Allocated to	Verification
SR-01	On application of rated kVA at 0.8 PF lagging in one step at rated speed, the alternator shall keep the transient voltage deviation at or above −15 % of rated voltage.	G3 value [S2][S5][S6]. The *full* single step is a mission assumption [A]: ISO applies −15 % to the engine's BMEP step, so SR-01 is stricter than the standard	Main machine, exciter, AVR	Analysis (here); Test (load bank)
SR-02	After the SR-01 step, the voltage shall return to, and stay within, ±1 % of its steady value within 4 s.	G3 t_U and the recovery band: "returns to and remains within the steady-state voltage tolerance band" [S2 3.32–3.33]; band width 2·δU_st [S3, 1993 ed. 8.8]	AVR, exciter	Analysis; Test
SR-03	On rejection of rated kVA at 0.8 PF, the transient voltage rise shall not exceed +20 %.	G3 [S5][S6]	AVR, exciter	Analysis; Test
SR-04	After SR-03, the voltage shall return within ±1 % within 4 s.	G3 [S5][S6]	AVR	Analysis; Test
SR-05	During a DOL start of the declared largest motor (132 kW pump), the dip shall stay at or above −20 %.	ISO 8528-5 has no motor-start clause [S2]; G4 means "by agreement" [S6]. −20 % is a common specifier limit [S8][S9]	Main machine, excitation	Analysis; Test
SR-06	For a bolted three-phase terminal fault, the alternator shall sustain at least 3 pu RMS current at 1 s.	Data-sheet practice for selective protection [A]; that shunt excitation does not sustain it is sourced [S7]	Excitation source	Analysis; Test
SR-07	No-load line-to-line voltage THD shall not exceed 5 %.	IEC 60034-1 clause 9.11 THD [S4]. The clause is verified; the 5 % value is the usual figure but was **not** seen in the standard	Winding, pole shape	Analysis (FEMM); Test
SR-08	Steady-state voltage shall stay within ±1 % from no load to rated load.	G3 [S5][S6]	AVR	Test

Derived requirements ([W] Principle 21.2, p. 437; allocation by synthesis, Principle 31.5, p. 669). The values are generated in §6.2 and §8.

ID	Derived requirement	From
DR-01	The main machine's X'd shall not exceed the value in the "largest X'd" table of §6.3, for the chosen excitation option.	SR-01, SR-05 via the X'd sweep
DR-02	The excitation system shall have its ceiling, TE and KC within the ranges for which §8.4 shows SR-01/02 met.	SR-01, SR-02
DR-03	The AVR shall be tuned so that the 5 % open-circuit set-point step overshoots by 10 % or less, while meeting SR-02.	SR-02 + a stability design rule [A] (IEEE Std 421.2-style performance indices)

Technical performance measures ([W] Principle 21.4, p. 442): X'd, X''d, T'do, the excitation ceiling, the SR-01 margin, the SR-02 margin and the largest motor at 20 %.

5. Architecture and interfaces ([W] ch. 26–27)



Architecture principles applied. Context, Principle 26.4 (p. 550); domain views, Principle 26.6 (p. 551); selection rationale, Principle 26.10 (p. 553), which is the trade study in §8.

ID	Interface	Definition	Owner
I-01	Voltage sensing	3-phase average RMS, filter TR = 10 ms [A]	AVR

ID	Interface	Definition	Owner
I-02	AVR supply	Shunt: ∝ terminal voltage, so the regulator ceiling scales with V_t (IEEE 421.5 AC8B note; AC7C/AC8C switch SW1 position A). PMG: independent of terminal conditions (SW1 position B) [S1]	Excitation
I-03	Exciter field	VR ≤ VRMAX (pu), exciter field time constant TE	Exciter
I-04	Rotating rectifier → main field	EFD = VE·FEX(IN), IN = KC·IFD/VE, VFE = (KE + SE)·VE + KD·IFD [S1]	Exciter, rotor
I-05	Shaft	Speed from the engine: PI governor with droop, actuator and combustion lags, boost-limited fuel [A]	Generating set (external)
I-06	LV bus	400 V, 50 Hz; block loads at 0.8 PF; motors start at PF 0.3–0.5 [S8]	Facility (external)
I-07	Test bench	OCC, field currents, regulation, build-up: measured by the sister bench project	Test

Fault containment ([W] Principles 26.16, p. 559, and 27.10, p. 596). The rotating rectifier is the single element whose failure is not visible from the stator side directly. §9 derives its signature on the stationary exciter-field side.

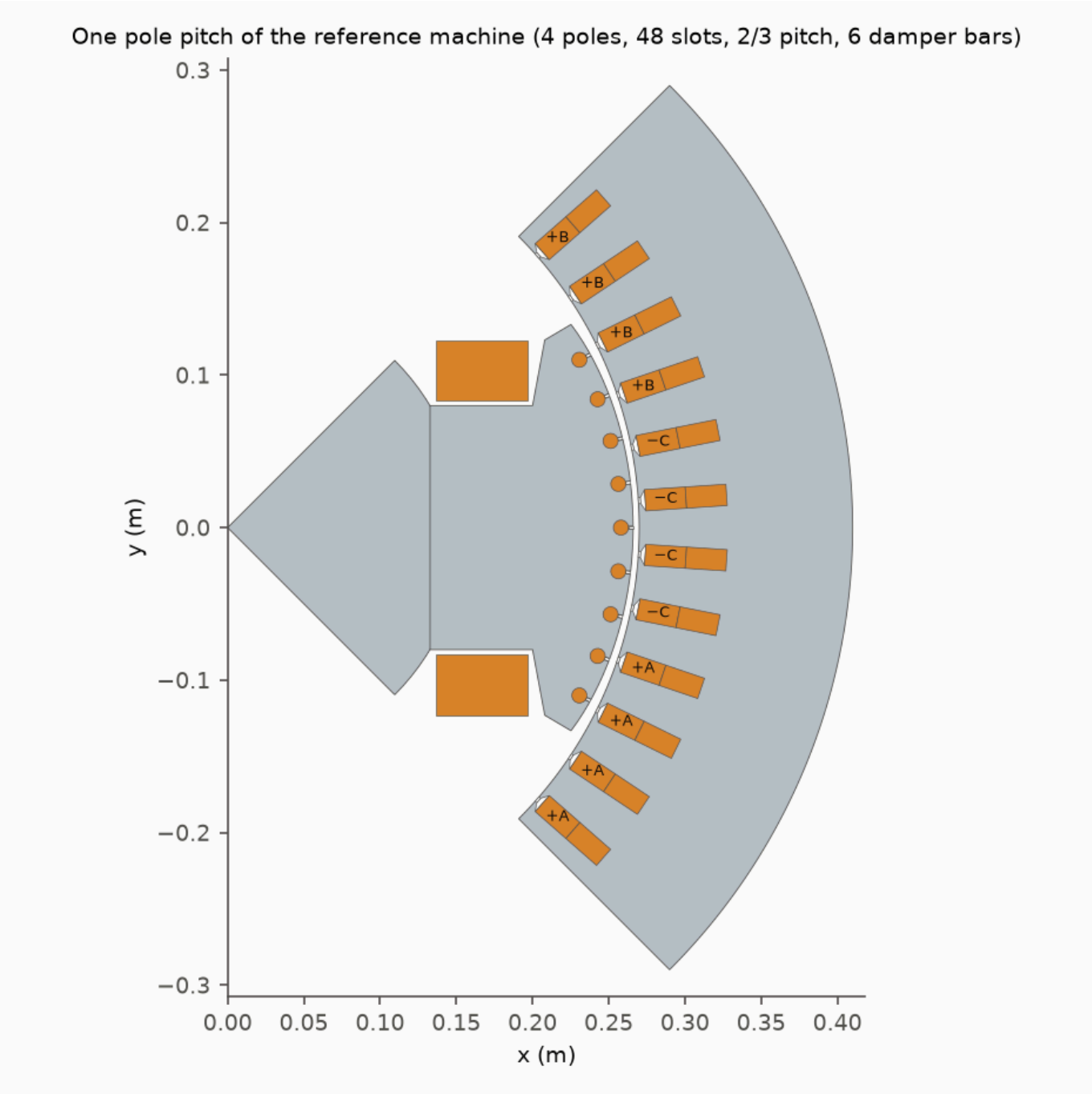
6. Design, performance budgets and margins ([W] ch. 31)

6.1 Reference design and its parameters

The reference machine is defined in `config/machine.yaml`:

- **Stator:** D = 540 mm, L = 430 mm, 48 slots, q = 4, double layer, 2/3 pitch (removes the 3rd harmonic), 12 series turns, 4 parallel paths.
- **Rotor:** salient 4-pole rotor with a cosine-shaped pole face (δ_0 = 4 mm), 120 field turns per pole, 9 damper bars per pole spanning 83 % of the pole arc (D17).

The analytic design (`bta/design.py`, method in `docs/design_notes.md`) gives the dimensions, the OCC and the reactances. The 2-D section (`bta/geometry.py`) is checked for fit: the field coil fits at fill 0.75 [A].



Parameters (generated):

Parameter (pu on 800 kVA)	Analytic (L1)	FEMM (L2)	Public data sheets 310–1150 kVA [S7]
Xd	2.386	2.196	2.44–2.94
Xq	1.091	1.001	1.90–2.48
Xl	0.117	0.117	—
X'd	0.265	0.256	0.146–0.19
X''d	0.145	0.207	0.11–0.13
X''q	0.131	0.160	0.24–0.37
T'do (s)	1.927	1.822	1.7–4.09
T'd (s)	0.214	0.213	0.08–0.097
T''do (s)	0.032	0.037	—
T''d (s)	0.017	0.030	0.012–0.019
T''qo (s)	0.398	0.229	—
Ra	0.011	0.011	—

FEMM results (generated):

Quantity	Value
No-load field current for 1 pu (FEMM)	28.8 A
Air-gap-line field current at 1 pu	26.9 A
Line-to-line THD, no load, unskewed	2.27 %
Phase THD (incl. triplen)	2.27 %
Largest line harmonics (% of fundamental)	5: 1.94 %, 7: 0.99 %, 11: 0.20 %, 13: 0.31 %, 19: 0.25 %, 25: 0.47 %
Field self-inductance Lff (4 poles)	1.798 H
Field–phase mutual Lafd	38.47 mH
SSFR fit RMS error, d (field open) / q	0.46 % / 0.23 %
SSFR low-frequency limit vs static solve (linear μr 2000 vs M-19 B-H), d / q	-4.3 % / -1.7 %
X''d: damper only (field open) / with field branch	0.345 / 0.207
T''d / T''do (circuit)	29.7 ms / 36.8 ms

Why FEMM and the analytic design differ (L1 vs L2, Principle 33.5, p. 718):

- **Xd, Xq (−8 %) and the air-gap-line field current (+7 %).** These are the same effect seen from two sides: FEMM's effective air gap is about 7 % larger. FEMM resolves three things the analytic model folds into a Carter factor and form factors for an ideal cosine pole:

- the stator slot openings;

- the nine 2 mm damper slits in the pole face (D16, D17);
- the finite iron permeability at the low test currents.

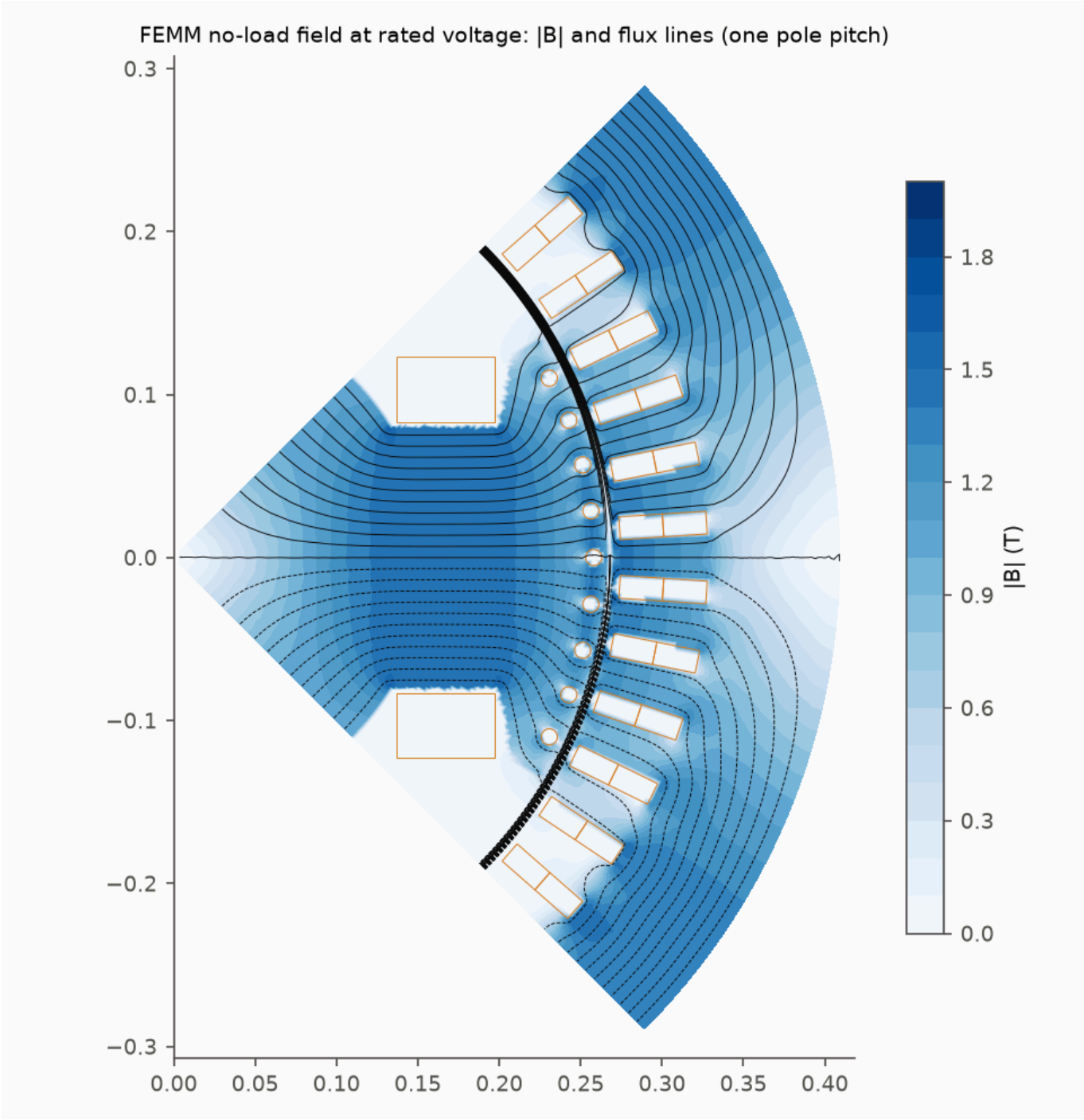
- **X'd (−3 %) and T'do (−5 %).** These are dominated by the field and stator leakage, which both levels compute similarly. The end-winding leakage is analytic in both. FEMM's Lff is 1.80 H against 1.90 H analytic.

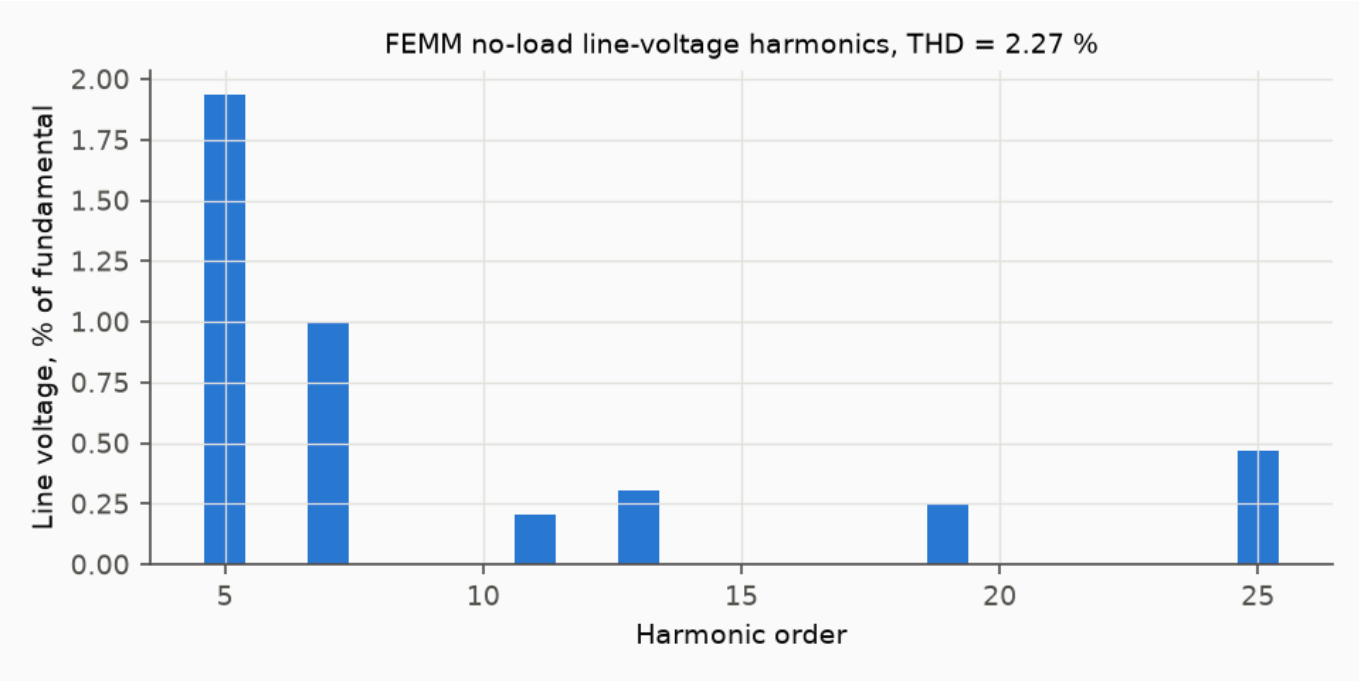
- **X''d (+43 %) and X''q (+22 %).** The analytic referral (`bta/design.py`, `damper()`) treats the bars as a cage spread evenly over the whole pole arc and ignores the slit leakage of each bar. In FEMM the 9 bars cover 83 % of the arc and each bar sits under its own slit, so less flux is screened. The 6-bar first run showed the same effect more strongly (X''d 0.242 pu, D17). The dynamic model uses FEMM's value, which gives the deeper and more conservative first dip.

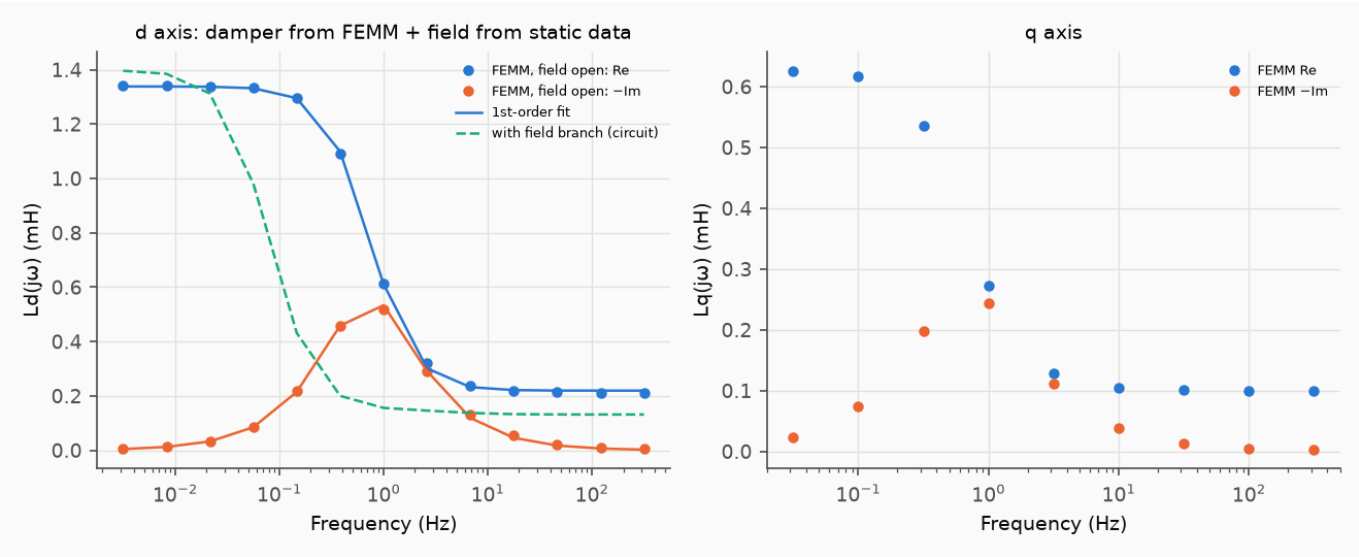
- **Field-branch combination.** The d-axis subtransient values combine the damper branch from the field-open SSFR with the field branch from the static solves, in the classical equal-mutual circuit (no Canay differential leakage). The fit errors are 0.46 % (d) and 0.23 % (q). The SSFR low-frequency limit is 4.3 % (d) and 1.7 % (q) below the static inductance, because the SSFR uses linear iron (μr = 2000) and the static solve the M-19 B-H curve.

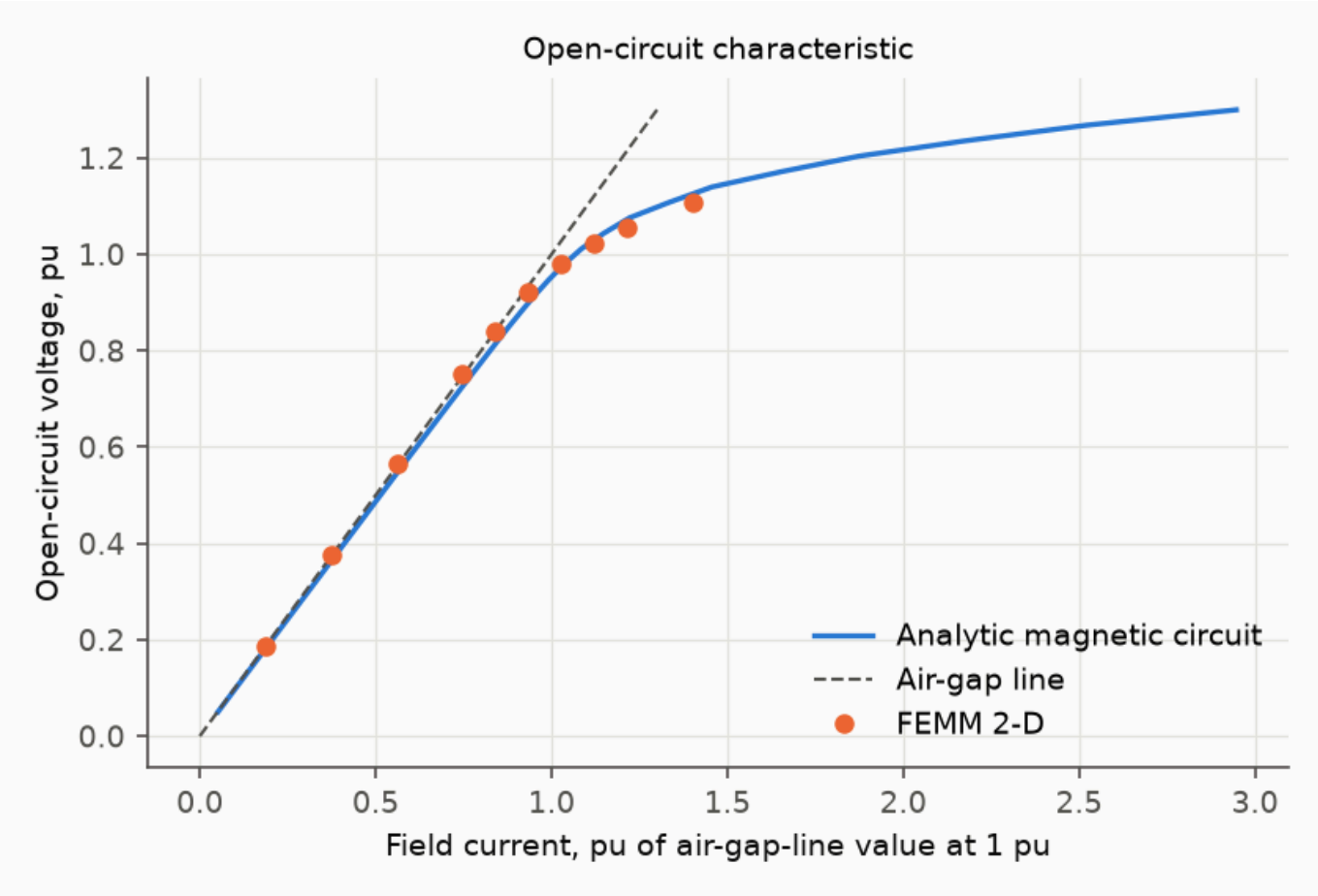
- **T''d.** It is 30 ms here against 12–19 ms on data sheets, because the FEMM end rings have zero resistance. Real ring resistance only shortens T''do and T''qo. Halving both (§8.4) deepens the SR-01 dip from −14.68 % to −15.05 %. The ideal ring is therefore slightly *optimistic*, and the 0.3-point margin of finding 3 may not survive real end rings. Measuring T''d (sudden short circuit, §10.3) is the first test to run.

- **Xq against public data.** Xq/Xd is 0.46 here against about 0.8 on the LV data sheets [S7]. This follows from the study design's pole arc (0.68) and wide interpolar gap. It enters the dip through X''q only weakly, and the data-sheet-class run in §6.3 brackets it.


FEMM no-load field at rated voltage: |B| and flux lines (one pole pitch)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
1.8
1.5
1.2
0.9
0.6
0.3
0.0
|B| (T)


FEMM no-load line-voltage harmonics, THD = 2.27 %
Line voltage, % of fundamental
Harmonic order
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
5
10
15
20
25


d axis: damper from FEMM + field from static data
FEMM, field open: Re
FEMM, field open: −Im
1st-order fit
with field branch (circuit)
Ld(jω) (mH)
Frequency (Hz)
q axis
FEMM Re
FEMM −Im
Lq(jω) (mH)
Frequency (Hz)



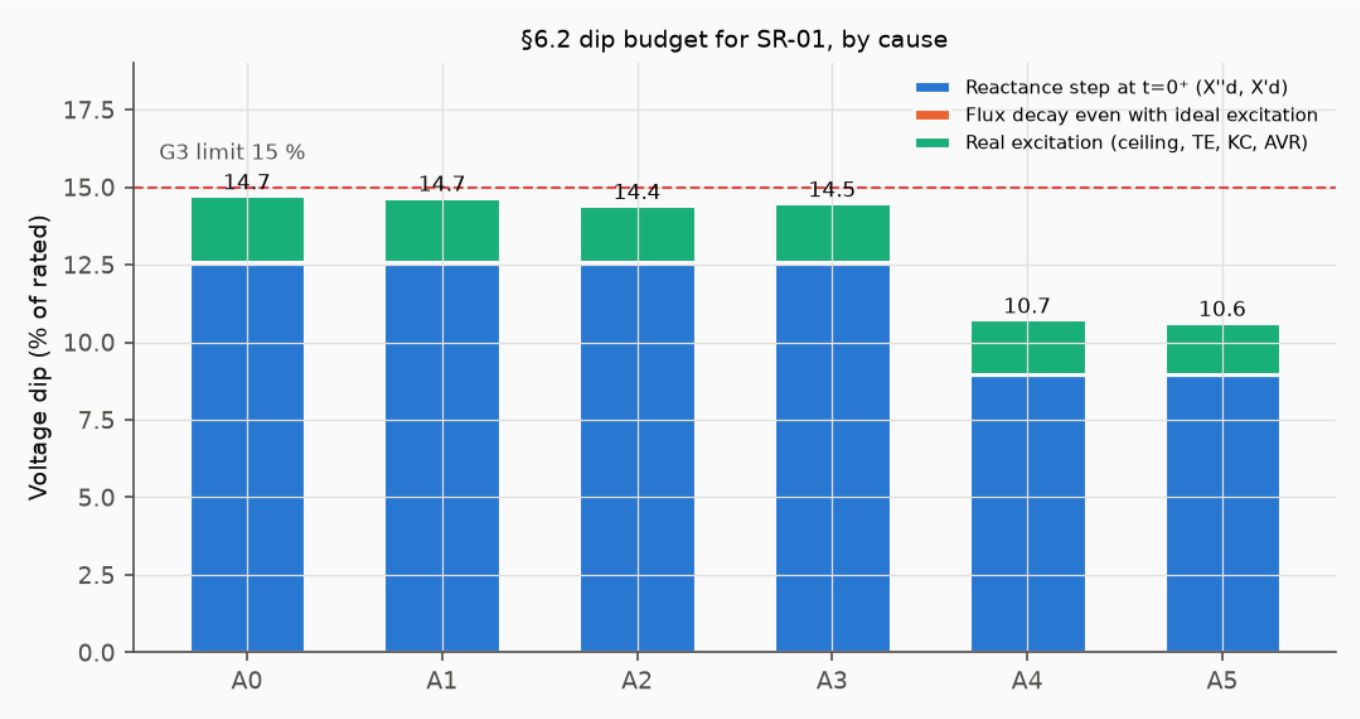
6.2 Voltage-dip budget for SR-01 (Principles 31.1–31.4, pp. 667–668)

The −15 % budget is apportioned to causes that the model can switch off one at a time (`scripts/run_studies.py`):

1. **Reactance step.** The dip at t = 0^+ with all rotor fluxes frozen. It is set by X''d, X''q and the load impedance, i.e. the main-machine design.
2. **Unavoidable flux decay.** The further dip even with an ideal excitation: no ceiling, TE = 5 ms, KC = 0.
3. **Real excitation.** The difference between that ideal and the real ceiling, TE, KC, AVR dynamics and supply.
4. **Engine speed.** Measured separately at a 50 % step, because a turbo-charged engine cannot take 100 % in one step. It is the extra dip in the generating-set run versus the constant-speed run.

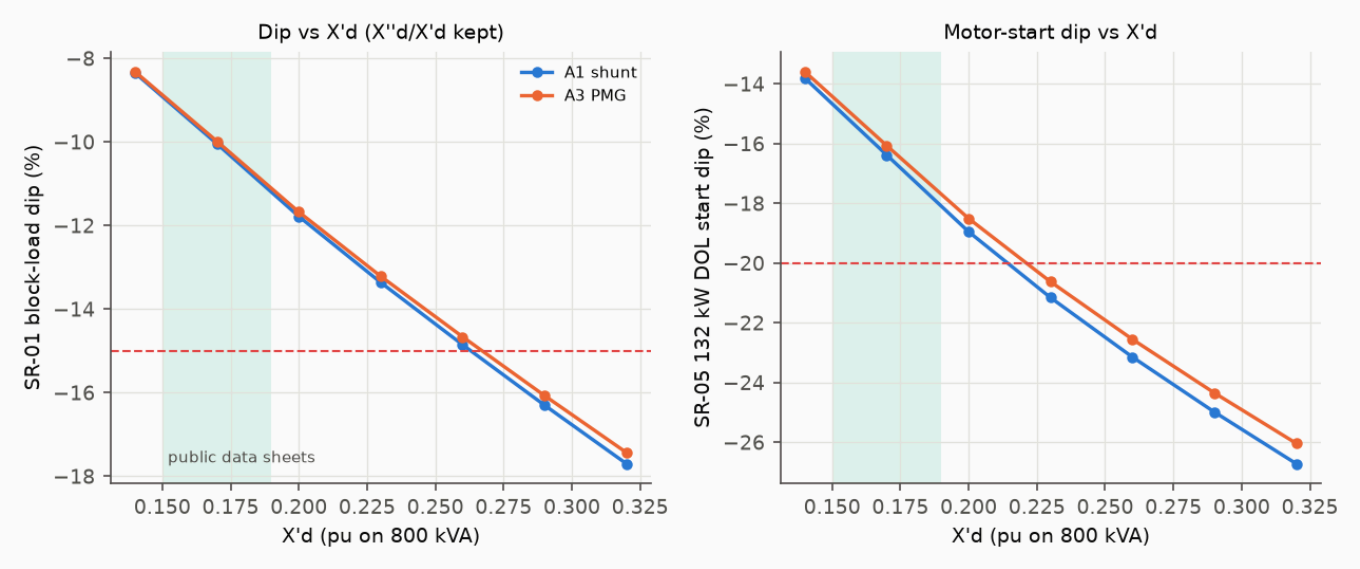
Results (generated):

	Reactance step $t=0^+$	Flux decay (ideal excitation)	Real excitation	Total dip	Margin to −15 %	Extra dip from engine speed (50 % step)
A0	12.60	-0.03	2.17	14.73	0.27	0.89
A1	12.60	-0.03	2.11	14.68	0.32	0.72
A2	12.60	-0.03	1.85	14.42	0.58	0.73
A3	12.60	-0.03	1.92	14.49	0.51	0.69
A4	8.98	0.00	1.76	10.74	4.26	0.74
A5	8.98	0.00	1.67	10.64	4.36	0.74



6.3 The deciding parameter: X'd

Excitation	Largest X'd meeting SR-01 (−15 %)	Largest X'd meeting SR-05 (132 kW, −20 %)
A1 shunt	0.263	0.214
A3 PMG	0.267	0.221



Robustness check against public data. A machine with data-sheet-class parameters (midpoints of [S7]) was run with the same excitation (generated):

Machine	Excitation	SR-01 dip	SR-02 t_U	132 kW dip
Data-sheet class (X'd 0.17)	A1 shunt retuned	-8.45 %	1.42 s	-16.2 %
Data-sheet class	A3 PMG retuned	-8.42 %	1.42 s	-15.7 %

7. Modeling and simulation ([W] ch. 33)

Fidelity (Principle 33.1, p. 706):

Level	Model	Resolves	Does not resolve
L1	Analytic design: magnetic circuit, permeances (`bta/design.py`)	Dimensions, OCC, Xd, Xq, X'd, X''d, T'do	Local saturation, slot ripple, true pole leakage
L2	FEMM 4.2, 2-D, one pole pitch with an anti-periodic air-gap element (`femm/`)	OCC, EMF waveform and THD, Ld and Lq, field self and mutual inductance, damper branch from a field-open SSFR (D15) → X''d, T''d, X''q	End effects (added analytically); skew; damper end-ring resistance (ideal rings)
L3	RMS dq dynamics (`bta/dynamics.py`): IEEE 1110 model 2.1, d-axis saturation from the OCC, AC8B/AC8C-type excitation, engine, induction motor	ISO 8528-5 metrics, ceilings, engine speed, motor start, fault current	First-cycle DC offset; waveform during the transient
L3b	Detailed rotating rectifier (`bta/rectifier.py`)	Diode commutation; open-diode and short-diode signatures	Exciter magnetic saturation and the exciter's own armature reaction

Model validity ("all models are wrong but some are useful", Principle 33.5, p. 718). Each model is checked against something independent of itself (`tests/test_models.py`, §10.2).

Interface choice. L2 feeds L3 through standard parameters. That is the same interface a test bench produces with IEC 60034-4-1 tests (§10.3), so the dynamic model can also be fed from measured data later.

8. Trade study: analysis of alternatives ([W] ch. 32)

Charter (Principle 32.3, p. 689). Find the cheapest change that makes the reference alternator meet SR-01 to SR-06.

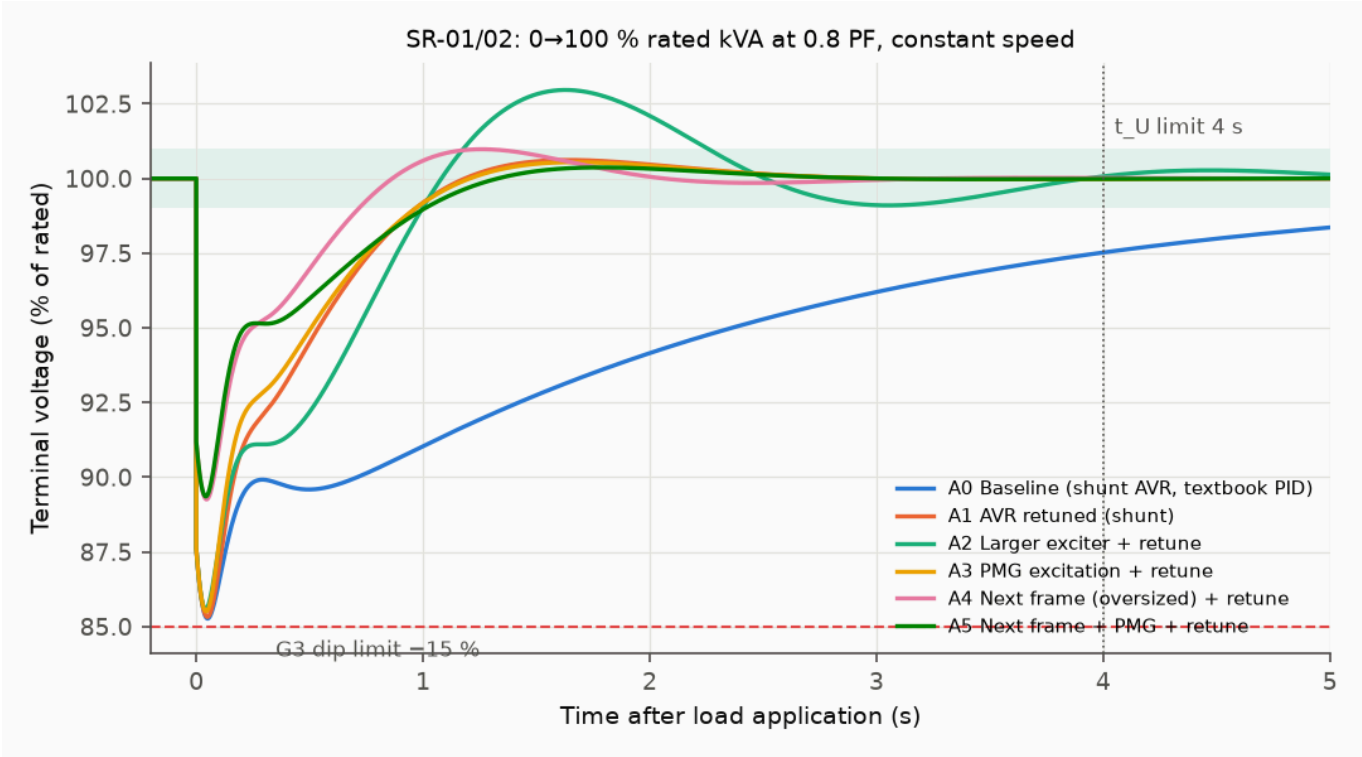
Decision factors and weights (normalized method, [W] §32.5, pp. 690–695):

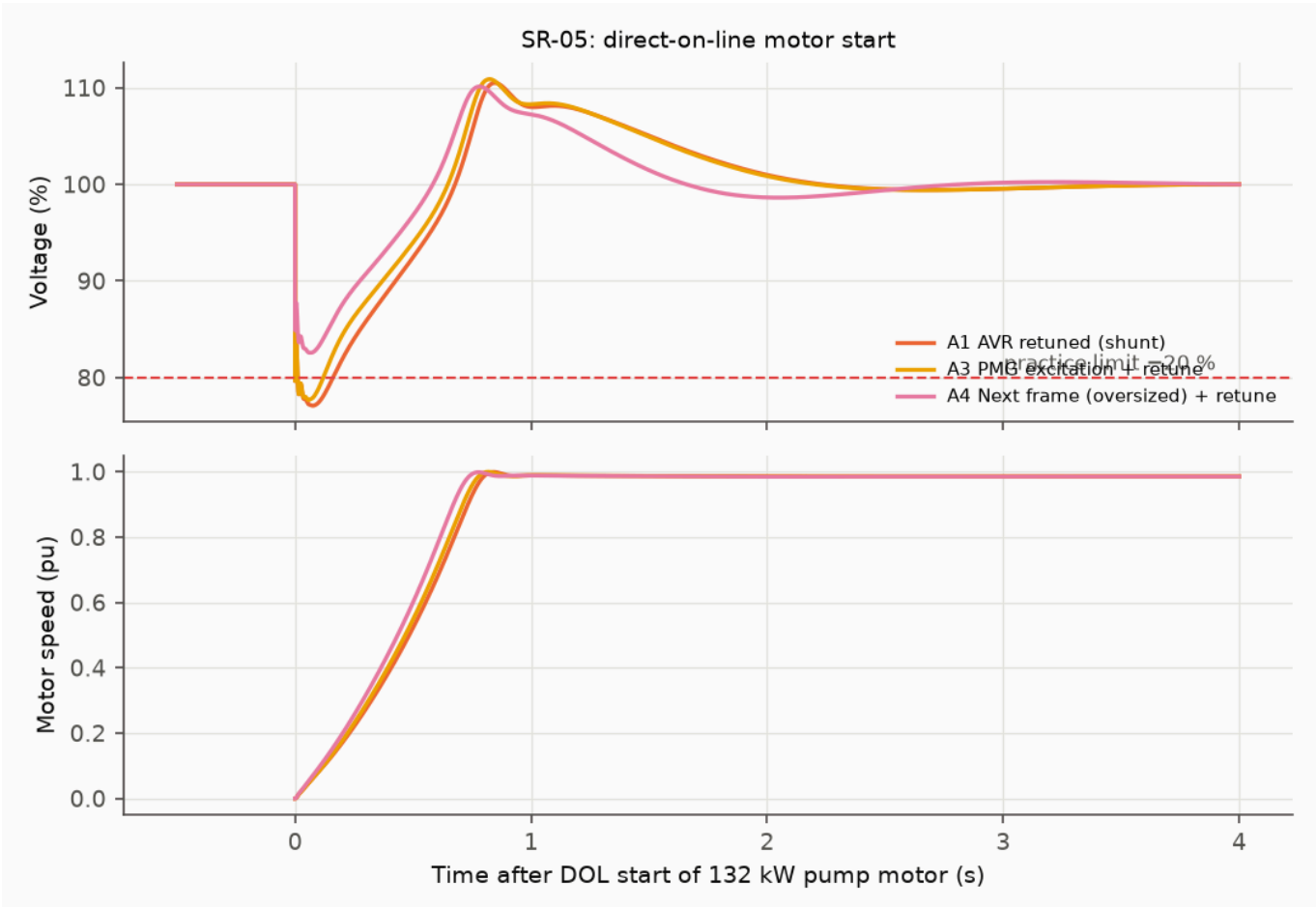
- Performance 50: SR-01 dip 15, SR-02 recovery 10, largest motor at 20 % 15, sustained fault current 10.
- Cost 25.
- Risk 15.
- Integration 10.

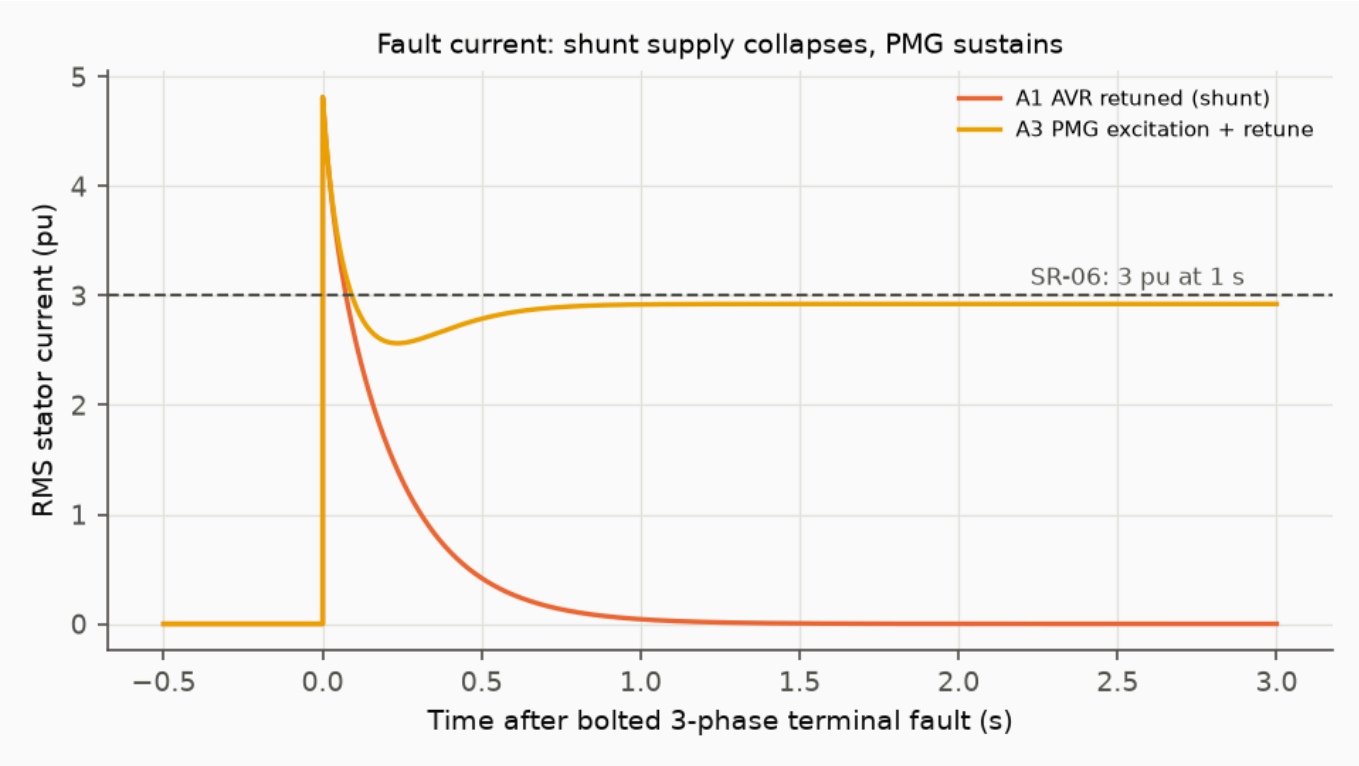
Utility is linear between the worst and the best alternative. Any "shall" that is not met disqualifies an alternative before scoring; that is why the scores are shown next to the gate results. Each retuned alternative is tuned on its own plant by a grid search over KP/KI/KD, with the open-circuit 5 % step overshoot held at 10 % or less (DR-03).

8.1 Results per alternative (generated)

	Alternative	KP/KI/KD	OC overshoot	SR-01 dip	SR-02 t_U	SR-03 rise	SR-04 t_U	SR-05 132 kW dip	Max motor @20 %	SR-06 I_sc at 1 s	Gates
A0	Baseline (shunt AVR, textbook PID)	20.7/10.0/4.6	0.0 %	-14.73 %	5.25 s	+17.0 %	5.11 s	-23.0 %	111 kW	0.04 pu	2/6
A1	AVR retuned (shunt)	30.0/60.0/8.0	6.0 %	-14.68 %	0.97 s	+16.9 %	1.85 s	-22.9 %	111 kW	0.04 pu	4/6
A2	Larger exciter + retune	15.0/60.0/8.0	6.5 %	-14.42 %	2.25 s	+17.3 %	3.27 s	-22.3 %	115 kW	0.04 pu	4/6
A3	PMG excitation + retune	30.0/60.0/8.0	8.3 %	-14.49 %	0.97 s	+16.9 %	1.85 s	-22.3 %	115 kW	2.92 pu	4/6
A4	Next frame (oversized) + retune	30.0/90.0/8.0	9.7 %	-10.74 %	0.72 s	+12.6 %	1.54 s	-17.4 %	151 kW	0.05 pu	5/6
A5	Next frame + PMG + retune	30.0/60.0/8.0	8.6 %	-10.64 %	1.01 s	+12.6 %	1.73 s	-17.1 %	156 kW	3.61 pu	6/6


SR-01/02: 0→100 % rated kVA at 0.8 PF, constant speed
Terminal voltage (% of rated)
Time after load application (s)
t_U limit 4 s
G3 dip limit −15 %
A0 Baseline (shunt AVR, textbook PID)
A1 AVR retuned (shunt)
A2 Larger exciter + retune
A3 PMG excitation + retune
A4 Next frame (oversized) + retune
A5 Next frame + PMG + retune


SR-05: direct-on-line motor start
Voltage (%)
Motor speed (pu)
Time after DOL start of 132 kW pump motor (s)
practice limit −20 %
A1 AVR retuned (shunt)
A3 PMG excitation + retune
A4 Next frame (oversized) + retune

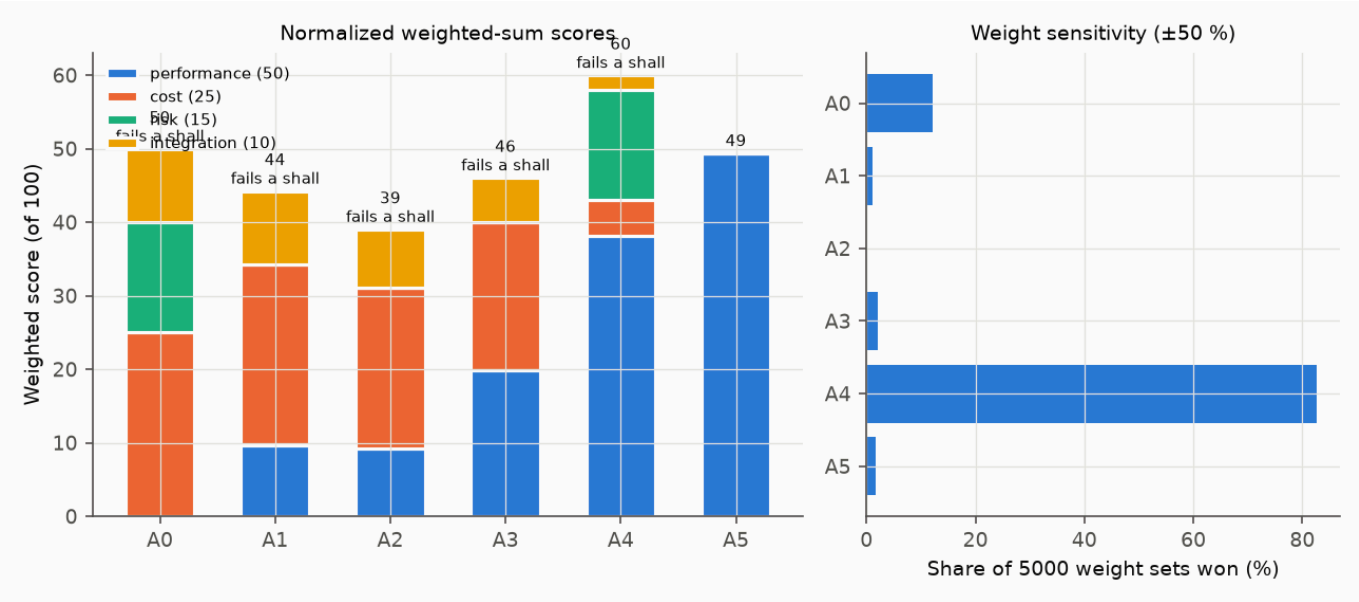


8.2 Requirement gates (generated)

Requirement	A0	A1	A2	A3	A4	A5
SR-01 dip	pass	pass	pass	pass	pass	pass
SR-02 recovery	**fail**	pass	pass	pass	pass	pass
SR-03 rise	pass	pass	pass	pass	pass	pass
SR-04 rejection recovery	**fail**	pass	pass	pass	pass	pass
SR-05 motor 132 kW	**fail**	**fail**	**fail**	**fail**	pass	pass
SR-06 sustained SC	**fail**	**fail**	**fail**	**fail**	**fail**	pass

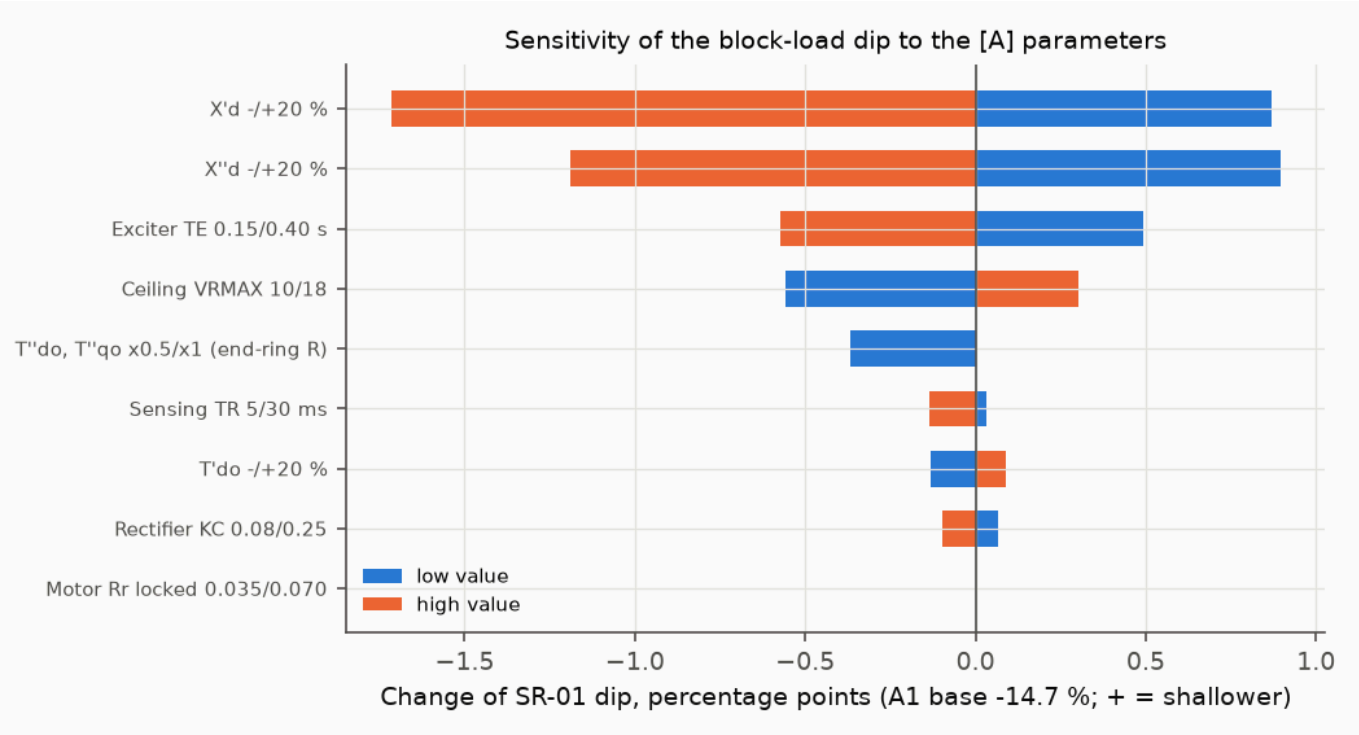
8.3 Scores and weight sensitivity (generated)

	Cost index [A/D]	Risk [A]	Integration [A]	Weighted score	Won (weights ±50 %)	Meets every shall
A0	1.00	9	10	50.0	12.2 %	no
A1	1.00	8	10	44.2	1.2 %	no
A2	1.04	8	9	39.0	0.0 %	no
A3	1.06	8	8	46.1	2.1 %	no
A4	1.25	9	6	60.0	82.7 %	no
A5	1.31	8	5	49.4	1.8 %	yes



8.4 Sensitivity to the assumed physical parameters (A1, generated)

Parameter [A] (low/high)	SR-01 dip low	high	t_U low	high	132 kW dip low	high
X'd -/+20 %	-13.81	-16.39	0.98	0.93	-21.6	-25.3
X''d -/+20 %	-13.78	-15.87	1.03	0.90	-22.0	-24.2
T'do -/+20 %	-14.81	-14.59	0.97	0.97	-23.2	-22.7
T''do, T''qo x0.5/x1 (end-ring R)	-15.05	-14.68	0.92	0.97	-23.6	-22.9
Exciter TE 0.15/0.40 s	-14.19	-15.25	1.00	1.63	-22.0	-24.1
Rectifier KC 0.08/0.25	-14.61	-14.78	0.95	0.99	-22.8	-23.1
Ceiling VRMAX 10/18	-15.24	-14.38	0.91	1.02	-24.3	-22.3
Sensing TR 5/30 ms	-14.65	-14.81	0.96	1.02	-22.9	-23.0
Motor Rr locked 0.035/0.070	-14.68	-14.68	0.97	0.97	-24.0	-21.2
A1 base	-14.68		0.97		-22.9	



8.5 Trade study report: recommendation (Principle 32.9, p. 698: recommendations, not decisions)

These are recommendations to the decision maker, not decisions (Principle 32.9).

1. **If SR-05 (a 132 kW DOL start within −20 %) and SR-06 (3 pu at 1 s) stay as written: A5**, the next frame with a PMG and a retuned AVR. It is the only alternative that meets all six shalls, at a cost index of 1.31.

- A4 scores highest and wins 83 % of the weight draws, but it fails SR-06. This is why the gates come before the scores.

2. **Challenge SR-05 before buying the frame** ([W] ch. 19, requirement validation). If the largest motor is held at 111 kW, or the 132 kW pump gets a soft starter, A3 meets every shall except SR-06. A3 is the PMG with a retuned AVR, at a cost index of 1.06. SR-06 is then met with a 4.4 % higher AVR ceiling. This saves about a fifth of the alternator cost index against A5.

3. **Study one more alternative before committing to the next frame: a base-frame redesign for X'd ≤ 0.21 pu.** Public 310–1150 kVA data sheets show X'd = 0.146–0.19 pu, and the data-sheet-class run passes SR-01 with 6.5 points of margin. X'd here is mostly leakage (field-pole leakage and stator slot leakage), so the levers are the pole and slot geometry, not the frame size. This alternative was not in the charter and is not scored.

4. **In every case**, retune the AVR with the DR-03 rule, since the textbook tuning fails SR-02. Also fit a rotating-diode monitor (1×/6× f_ex): on the base frame a single open diode is enough to break SR-01 without any alarm.

9. Reliability: FMEA ([W] ch. 34)

Definitions (MIL-HDBK-470A as quoted in [W] §34.3.8, pp. 761–762):

- failure mode: "the consequence of the mechanism through which the failure occurs";
- effects at three levels: local, next higher level, end.

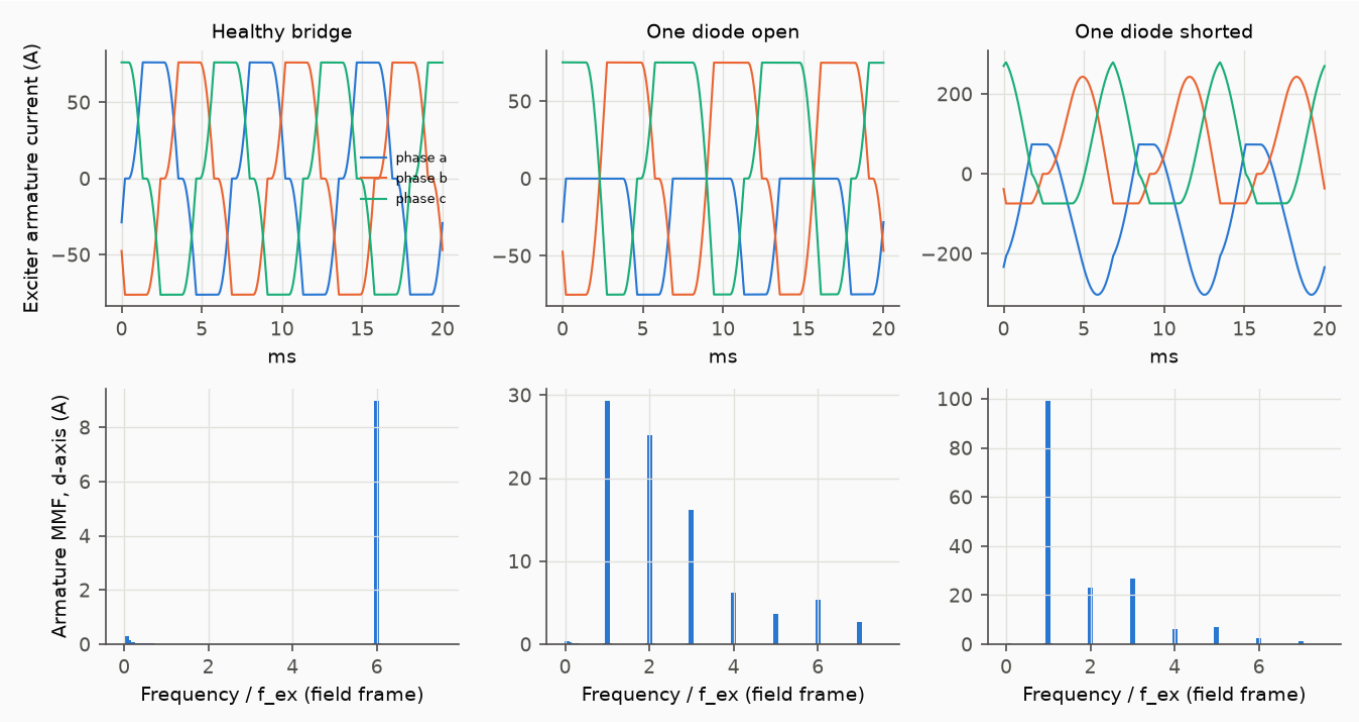
Severity (S), occurrence (O) and detection (D) are rated 1–10 [A]; RPN = S·O·D. Items marked ▲ are quantified by simulation.

ID	Item / failure mode	Local effect	Next higher level	End effect	Detection	S	O	D	RPN	Mitigation
FM-01 ▲	Rotating diode open	One phase of the exciter armature conducts half-wave; DC component in the armature currents	Rectifier capability drops (factor in the table below); AVR raises the exciter field current	Set still runs; deeper and slower transients (below); hidden failure	Exciter field current component at f_ex [D], [S11] [S13]	5	3	7	105	Diode monitor on exciter field current (1×f_ex / 6×f_ex ratio, [S12]); routine bench check
FM-02 ▲	Rotating diode short	Exciter armature line-to-line short each cycle	Armature current ×2.9, capability about halved; AVR at or near ceiling	Exciter overheating, then loss of excitation; collapse on the next large step	Same signature, much larger [D], [S12]	8	2	6	96	Same monitor with a trip; varistor/surge suppressor on the rectifier
FM-03	AVR sensing lost (open fuse or wire)	AVR sees 0 V	Regulator drives to ceiling	Over-voltage on all loads	Over-voltage relay	9	2	3	54	Sensing-loss detection in the AVR; over-excitation protection
FM-04	PMG failure (A3/A5 only)	No AVR supply	Excitation lost	Voltage collapse, set trips	Under-voltage relay	8	1	2	16	PMG fault alarm; robust PMG mounting
FM-05	Loss of remanence (shunt only)	No build-up	Set cannot excite	Failed start (mission failure)	Start sequence time-out	7	2	2	28	Field flashing; PMG option removes it

ID	Item / failure mode	Local effect	Next higher level	End effect	Detection	S	O	D	RPN	Mitigation
FM-06	Damper bar crack	Damper resistance up	X''d, T''d change	Slightly deeper first dip; local heating at the crack	Rotor inspection	4	2	8	64	Bar/ring brazing inspection
FM-07	Field inter-turn short	Lower field ampere-turns per amp; unbalanced pole MMF	Higher field current at load; magnetic pull imbalance	Vibration, field overheating	Field current vs load trend; vibration	6	2	6	72	Routine field resistance and impedance (pole-drop) test
FM-08	Exciter field open	No exciter output	Excitation lost	Collapse	AVR output check	8	1	2	16	Exciter field resistance check in the routine test
FM-09	Main stator insulation fault	Earth fault	Protection trips	Outage	Earth-fault relay; IR test	9	2	3	54	IR/PI and HV tests (routine test)
FM-10	Over-tuned AVR (configuration)	Oscillatory voltage	Fails SR-02/04; can hunt with the engine	Flicker, trips	OC step test	5	3	3	45	DR-03 tuning rule; OC step test in commissioning

Diode failure, quantified (generated; `bta/rectifier.py` + dynamic model with the rectifier capability factor):

Rectifier state	Mean DC voltage (V)	Capability factor	Max armature current vs healthy	Field-frame MMF at f_ex (A)	at 6 f_ex (A)	ratio 1/6	SR-01 dip	SR-02 t_U	132 kW dip
Healthy	51.1	1.000	1.00	0.01	8.98	0.002	-14.68 %	0.97 s	-22.9 %
One diode open	39.1	0.765	1.06	29.31	5.30	5.530	-15.02 %	1.11 s	-23.7 %
One diode shorted	26.3	0.515	2.90	99.20	2.36	41.946	-15.67 %	1.54 s	-25.3 %



The detailed rectifier model (healthy case, negligible armature resistance) reproduces the classical six-pulse result with commutation overlap, $V_{dc} = (3\sqrt{2}/\pi)\cdot V_{LL} - (3/\pi)\cdot X_c \cdot I$, within 1 % (test `test_healthy_bridge_dc_voltage_matches_commutation_formula`). A failed diode puts a component at the exciter frequency f_{ex} into the field-frame armature MMF, and the $1\times f_{ex}$ / $6\times f_{ex}$ ratio rises sharply. This matches the detection principle of [S12] and the field-current signatures reported in [S11][S13]. Only frequencies and relative

changes are claimed: the amplitude reaching the exciter field current depends on the field circuit and the AVR, which were not modelled.

10. Verification and validation ([W] ch. 13, 28)

Verification asks "built right to the spec?" (Principle 13.3, p. 275). Validation asks "right system for the need?" (Principle 13.4, p. 275). Following Principles 13.11–13.12 (p. 283) and 28.3 (p. 608), one test should verify several requirements.

10.1 Verification matrix

Req.	Method in this study	Evidence	Method on a real machine
SR-01, SR-02	Analysis (L3)	§8.1, `results/aoa_femm.json`	Load-bank test: 100 % 0.8 PF step on the bench (one test verifies both)
SR-03, SR-04	Analysis	§8.1	Same test, load rejection
SR-05	Analysis	§8.1, max-motor search	Motor-start test or a low-PF load-bank simulation (ISO 8528-3:2020 Annex A.3 "motor starting loads" [S3])
SR-06	Analysis	§8.1	Sustained three-phase short-circuit test
SR-07	Analysis (FEMM L2)	§6.1	Waveform measurement at no load (IEC 60034-1 cl. 9.11.3 [S4])
SR-08	Not analysed (steady state)	—	Regulation test (sister bench check "AVR regulation")

10.2 Model verification (each against an independent result; `tests/test_models.py`)

Check	Independent reference	Result
Winding factors, 2/3 pitch kills the 3rd harmonic	Closed form	pass
Steady states are steady (no load, 0.8 PF, unity PF)	Definition	pass
Voltage at $t = 0^+$ of a step	Closed form with frozen fluxes: $V = 1/\|1 + Z''\cdot Y\|$	within 0.01 pu
Induction-motor model	Textbook equivalent circuit at 4 slips	exact (1e-9)
Shunt collapses / PMG sustains fault current	Data-sheet statement [S7]	reproduced
Rectifier average voltage ($R \to 0$)	Six-pulse commutation formula	within 1 %
Rectifier fault ratio 1×/6× f_ex rises with the fault	Detection principle [S12]	reproduced
L1 analytic vs L2 FEMM	Independent method	§6.1 table
SSFR low-frequency limit vs static Ld, Lq	Two FEMM routes (time-harmonic linear iron vs magnetostatic B-H)	§6.1 FEMM table

10.3 Bench test plan (links to the sister project *alternator-routine-test-station*)

The sister project simulates the routine-test bench. Its 800 kVA "medium" frame has the same rating as this machine. This model provides the expected values that bench checks compare against, and it adds the type tests needed to measure the model inputs:

Test	Standard clause [S10]	Measures	Model input or output
Open-circuit characteristic	IEC 60034-4-1 6.4 / 7.2.1	OCC, no-load field current	Saturation table; bench check "no-load field current"
Sustained short-circuit characteristic	6.5 / 7.2.1	Xd (unsaturated) with the OCC	Xd
Sudden three-phase short circuit	6.11 / 7.1.2, X'd 7.3.1, X''d 7.4.1, T'd 7.16.1, T''d 7.18	Transient and subtransient quantities	X'd, X''d, T'd, T''d (validates L2)
Low-slip test	6.10 / 7.5.2	Xq	Xq
Standstill frequency response	6.27 / 7.28.3, Annex B	Operational inductances Ld(jω), Lq(jω)	Direct comparison with the FEMM SSFR curve
DC decay at standstill	6.14 / 7.1.4	Time constants	T'do

Test	Standard clause [S10]	Measures	Model input or output
Load-bank transient test	ISO 8528-5 / -3	δU_dyn, t_U	Validation of L3 (SR-01/02)

[S10] notes that brushless machines "require special effort for some of the tests": there is no slip-ring access to the main field. So the SSFR and sudden-short-circuit tests are type tests on an instrumented rotor, not routine tests.

10.4 Validation

Validation means operational test and evaluation by the user (Principle 13.14, p. 289). It needs the generating-set packager's own load-acceptance test with the real engine, recorded against UC-1 to UC-3. The engine model here is generic [A]. Its effect (§6.2, last column) is therefore a sensitivity, not a validated prediction.

11. Assumed versus sourced, and limits

Sourced [S]. The following come from sources:

- ISO 8528-5 G3 values (through two white papers) and the recovery-band definition.
- IEEE 421.5 model structures, the FEX function and the shunt/PMG supply switch.
- The IEC 60034-4-1 test clauses.
- The public data-sheet parameter ranges.
- Motor-starting practice values.
- The qualitative behaviour of shunt vs PMG excitation.
- The diode-failure detection principles.
- The FEMM semantics used for damper bars and the shorted field [T7].

Derived [D]. Everything computed by the code in this repository.

Assumed [A], and where it matters:

- **Geometry.** The machine is a study design, not a product. Its X'd (§6.1) sits **above** the public data-sheet range of 0.146–0.19. §6.3 shows the conclusion depends on exactly this, which is why the X'd threshold, not the specific machine, is the main result.
- **Excitation parameters** TE, KC, KD, SE points and the 6 A AVR forcing current behind VRMAX: §8.4 varies them.
- **Engine model** (droop, lags, turbo-lag fuel limit): it only enters §6.2's engine column.
- **Cost, risk and integration scores:** varied in the weight Monte Carlo. The cost of the next frame is derived from the active-volume ratio.
- **The 132 kW motor and its parameters:** the starting PF follows from the parameters (0.34 at locked rotor, within the 0.3–0.5 of [S8]).

Limits:

- 2-D FEMM without skew or end effects; the end-winding leakage comes from the analytic model.
- Damper end rings are ideal in FEMM. §8.4 bounds the effect: halving T''do and T''qo breaks SR-01 (−15.05 %).
- RMS dynamics only, so no first-cycle offset.
- One design point.
- The textbook equations [T1]–[T5] are derived in `docs/design_notes.md` but their equation numbers were not re-checked against the books.

12. Decision log

#	Decision	Alternatives considered	Rationale
D1	Reference machine 800 kVA, 400 V, 4-pole	Smaller or larger frames	Same rating as the sister bench project's "medium" frame, so this model can give that bench its expected values
D2	2/3-pitch double-layer winding, q = 4	5/6 pitch	Removes the 3rd harmonic, the usual choice for LV generating sets with loaded neutrals
D3	RMS (phasor-dynamics) fidelity for L3	Full EMT (stator transients, diode switching)	ISO 8528-5 limits apply to the RMS envelope. EMT would add first-cycle detail at about 100× the runtime ([W] Principle 33.1)
D4	IEEE 421.5 AC8B structure with the AC8C SW1 supply switch	AC1A, AC5A	PID AVR + brushless exciter + FEX matches a digital LV AVR. SW1 is the shunt/PMG question of the trade study [S1]
D5	VRMAX = 14 pu (derived from an assumed 6 A AVR forcing current)	8 pu (first estimate)	The first estimate came from the exciter-field voltage, not current. 14 pu reproduces the about 3 pu PMG fault current that data sheets show
D6	Requirement SR-01 applied to a full 100 % step	ISO's BMEP-based step	Mission assumption for a critical-facility standby set. It is stricter than ISO, stated as [A]
D7	AVR tuning by an auditable grid search with an OC-overshoot ≤ 10 % rule	Optimiser; pole cancellation only	A grid is reproducible and shows the trade-off. Pole cancellation alone left a slow integral tail under load
D8	Rotor geometry changed to pole body 160 mm, 15 mm² edge-wound strip	180 mm body, 18 mm² wire	The 2-D section showed the 18 mm² coil did not fit between the poles (`bta/geometry.check_section`)
D9	FEMM rotor rotation by an anti-periodic air-gap element	Re-meshing per position	No remeshing noise in the waveform; one model for all positions [T7]
D10	FEMM solver run visible, at above-normal priority	Minimised window	A minimised solver was throttled by the OS on a busy workstation (observed 10 % CPU duty)
D11	Next-frame alternative designed with the same code, FEMM correction carried over	Scale per-unit values by kVA	Keeps the comparison like-for-like and traceable to geometry
D12	Cite only three neutral public data sheets	Four sheets	Keeps company and product names of prospective employers out of the repository
D13	THF not computed	Compute with a remembered weighting table	The THF weighting table could not be verified, and the current IEC 60034-1 contents list clause 9.11 as THD, not THF [S4]. So only THD is claimed
D14	SSFR in FEMM with linear iron, μr = 2000 [A]	Nonlinear iron in the time-harmonic solver	SSFR is a small-signal test at a few % of rated current. The nonlinear harmonic solve needed minutes per point with no physical gain

#	Decision	Alternatives considered	Rationale
D15	SSFR with the field OPEN and the bars joined by ideal end rings (bar connection: see D17). The field branch comes from the static FEMM solutions (Lff, Lafd) and the design resistance, combined in the d-axis circuit	Field modelled as one solid shorted loop	The first SSFR run with a solid-loop field gave Ld = 0.135 mH flat above 1 Hz, and X''q ≈ X'q. The solid loop crowds eddy currents to the coil edges and screens like an extra damper, which a stranded coil cannot do. (A remark in an earlier draft that unconnected bars left the q-axis damper inactive came from the closed-bridge geometry of D16 and is withdrawn; see D17.) The raw first run is kept as `femm/out/prelim_closed_damper_slots/ssfr_field_solid_loop.csv`
D16	Damper bars in semi-open slots (2 mm slit to the pole surface)	Bars fully enclosed under a 3 mm iron bridge	With a closed bridge and linear iron, the bridge carried the flux around the cage: the bar-to-bar flux difference was about 1000× smaller than the pole flux, and X''d ≈ X'd. Real amortisseur slots are semi-open, and the analytic damper permeance already assumed the 2 mm opening. The closed-slot results are kept in `femm/out/prelim_closed_damper_slots/`
D17	9 damper bars per pole (8 pitches of 0.85 slot pitch = 83 % of the pole arc), each bar shorted by an ideal continuous end ring (FEMM block with no circuit)	6 bars (52 % of the arc); all bars of the pole in one zero-net-current parallel circuit	The full FEMM run with 6 bars gave X''d = 0.242 pu against X'd = 0.257 pu and X''q = 0.90 pu: the outer half of the pole face had no damper, while the analytic model treats the cage as spread over the whole arc. The zero-net constraint is also wrong for the q axis: there the bar currents of one pole have the same sign and return through the ring to the next pole. Under anti-periodic symmetry an ideal continuous ring forces the common bar voltage to zero, which is exactly a no-circuit block ($J = -j\omega\sigma A$). A quick check at 100 Hz (2-D inductance, `femm/out/damper_check_6v9_bars.txt`): d axis 0.531 → 0.190 mH (6 → 9 bars, same for both connections); q axis 0.388 mH (parallel circuit) vs 0.078 mH (no circuit). The 6-bar results are kept in `femm/out/prelim_6bars_zero_net/`

13. References

- **[W]** C. S. Wasson, *System Engineering Analysis, Design, and Development: Concepts, Principles, and Practices*, 2nd ed., Wiley, 2016. Chapters 4–7, 13, 19–22, 26–28, 31–34.
- **[S1]** IEEE Std 421.5-2016. Read through public implementations:
- AC8B: CIM ExcIEEEAC8B, https://acs.pages.rwth-aachen.de/public/cim/cimpy/cimpy.cgmes_v2_4_15.ExcIEEEAC8B.html
- AC8C, switch SW1: MathWorks, https://www.mathworks.com/help/sps/ref/smac8c.html
- IN, FEX: GridKit EXAC1, https://gridkit.readthedocs.io/en/latest/GridKit/Model/PhasorDynamics/Exciter/EXAC1/README.html
- **[S2]** ISO 8528-5:2018. Contents (clause 15) and definitions 3.32–3.33 from the public sample, https://cdn.standards.iteh.ai/samples/71009/8a656586f1e44760a70afec399718590/ISO-8528-5-2018.pdf
- **[S3]** ISO 8528-3:2020 (Annex A.3) and ISO 8528-3:1993 (8.8 tolerance band), public samples at cdn.standards.iteh.ai (samples 77444, 15759).
- **[S4]** IEC 60034-1:2022 contents, clause 9.11 "THD for synchronous machines" (public preview, webstore.iec.ch / iteh sample 103428).
- **[S5]** Manufacturer white paper, "ISO 8528-5 and Generator Transient Performance" (2022 edition values), https://techcomm.kohler.com/techcomm/pdf/ISO%208528-5%20and%20Generator%20Transient%20Performance_WP.pdf
- **[S6]** Manufacturer white paper on generating-set performance classes (2018 edition values), https://mart.cummins.com/imagelibrary/data/assetfiles/0058629.pdf
- **[S7]** Three public LV alternator data sheets, 310–1150 kVA, 400 V 50 Hz (reactances, time constants, motor-starting curves, self-excited short-circuit note); URLs and values in `docs/sources_datasheets.md`.
- **[S8]** Manufacturer white paper TP-103, "Sizing generators for motor starting" (starting PF 0.3–0.5, NEMA code letters), https://resources.kohler.com/power/kohler/industrial/pdf/Sizing_Generators_for_Motor_Starting_WP.pdf

- **[S9]** L. A. Bey, "Sizing gensets for large motor starting", EC&M, https://www.ecmweb.com/basics/article/20885994/sizing-gensets-for-large-motor-starting
- **[S10]** IEC 60034-4-1:2018. Contents, scope and clause numbers from the public preview, https://assets.vde-verlag.de/iec-normen/preview-pdf/info_iec60034-4-1%7Bed1.0%7Db.pdf
- **[S11]** M. A. McArdle, D. J. Morrow, "Noninvasive detection of brushless exciter rotating diode failure", IEEE Trans. Energy Conversion, 2004, doi:10.1109/TEC.2003.822325 (abstract).
- **[S12]** US Patent 9,910,083 B2: rotating diode failure detection using the ratio of exciter field current harmonics.
- **[S13]** Tian et al., brushless exciter diode failure, SDEMPED 2021 (abstract): a failed diode raises the exciter field current and unbalances the armature currents.
- **[S14]** Dealer application note, "Motor start concerns": a PMG does not significantly reduce the initial dip, https://www.toromontpowersystems.com/pdf/ConsultantsCorner/Motor%20Start%20Concerns.pdf
- **[T1]** J. Pyrhönen, T. Jokinen, V. Hrabovcová, *Design of Rotating Electrical Machines*, 2nd ed., Wiley, 2014.
- **[T2]** I. Boldea, *Synchronous Generators*, 2nd ed., CRC Press, 2016.
- **[T3]** P. Krause et al., *Analysis of Electric Machinery and Drive Systems*, 3rd ed., Wiley-IEEE, 2013.
- **[T4]** P. Kundur, *Power System Stability and Control*, McGraw-Hill, 1994 (§3.8 saturation, §4.2 parameter relations, §7.2 induction motors).
- **[T5]** IEEE Std 1110-2019, guide for synchronous generator modeling practices (model 2.1 structure).
- **[T6]** MIL-HDBK-470A, as quoted in [W] §34.3.8.
- **[T7]** D. Meeker, *FEMM 4.2 User's Manual*, §2.2.7 "Circuit properties" (pp. 32–33: parallel circuits; conductors without a circuit are "shorted together at infinity"), and the Lua command reference for anti-periodic air-gap elements (p. 90–91). The copy installed with FEMM.